

MASTER'S COMPREHENSIVE EXAMINATION

Part II Applied (2 hours) December 2nd, 2017.

Instructions: Attempt any three out of the following four questions.

Mention which three you would like to have graded. Open book and open notes.

1. A researcher seeks to compare 4 different diets on the egg production of chickens (hens). Each diet is of a mixture of one Energy feed and one Protein Feed as follows

- Two Possible Energy Feeds Are: Corn(C), or Barley(B)
- Two Possible Protein Feeds: Soybean Meal (SM) or Cottonseed Meal (CM)
- The 4 possible diets are Diet-1: (C,SM) ; Diet-2: (C,CM) ; Diet-3: (B,SM) ; Diet-4: (B,CM)

The response variable for each chicken is the egg production - number of eggs in a month.

A random sample of 12 chicken farms were picked from a larger population of chicken farms. Then 3 of the 12 farms were assigned (completely at random) to Diet-1; 3 farms to Diet-2; 3 farms to Diet-3; and 3 farms to Diet-4. On each farm, 20 chickens were fed the assigned diet, and egg yield was recorded for each chicken. (There are a total of 240 chickens in this experiment).

Now Answer the Following

A. For the factors: Energy Feed, Protein Feed, Farms, Chickens, and any interactions, state which involve fixed effects, and which involve random effects?

B. For each pair of factors, state which are crossed, and which are nested.

C. Show the ANOVA Table showing Source of Variability, Degrees of Freedom, Expected Mean Squares, and show which F Tests to take. .. (Assume the normality and homogeneity assumptions of ANOVA are satisfied.). The researcher is specifically interested in the effects on egg production of Energy Feed, Protein Feed and Their Interaction. Your ANOVA table should show what is the numerator and denominator to test each of these factors.

2. Dr. Jones is testing a new home delivered education program to increase kindergarten test scores. He will randomly chose 50 independent children to have his program given to them by their parents and then compare the results to 50 other independent children who are not given the program. From normative data, typical education test scores without the intervention have a standard deviation of 15 units and are normally distributed. Higher test scores are better. He will test the null hypothesis that the intervention has no effect on test scores with a two sided $\alpha = 0.05$. Let Δ represent the true amount of units on average that the intervention improves test score in the population.

- A. Assume the intervention has no effect on the standard deviation of a test score and give the power to detect a difference of $\Delta=14$ under H_A .
- B. Now further assume that after the study is done that 21 of the 50 parents who were supposed to give the program to their child did not, so the child had no benefit,
- Assuming the 21 parents cannot be identified and all 50 children randomized to receive Dr. Jones program must be analyzed together, will power to detect $\Delta = 14$ under H_A be more or less than what is calculated above for Part A? Explain why.
 - If the 21 parents can be identified, will excluding their children from the 50 and only comparing the 29 who delivered the program to the 50 controls typically improve power over analyzing all 50 together? Explain why or why not.

3. A commuter uses Google Maps to estimate the travel time by car to get from downtown New Brunswick to 34th Street and 1st Avenue in New York City. The departure times can be 6 AM, 7 AM, or 8 AM. Google maps gives two alternative routes: either take the Lincoln Tunnel, or the Holland Tunnel. Assume that the following are Google maps times (in minutes):

	Lincoln Tunnel			Holland Tunnel		
	6AM	7AM	8AM	6AM	7AM	8AM
Monday November 13	79	100	180	80	110	168
Tuesday November 14	90	120	172	95	122	165
Wednesday November 15	90	115	185	107	128	171
Thursday November 16	87	150	180	90	130	165

Assume that travel times between days of week are independent of each other. However travel times within a day are not independent. Also travel times using Lincoln tunnel and Holland tunnel on same day and starting time are not independent of each other (both routes use the NJ turnpike for part of trip).

- Is there statistically significant evidence (at the 0.05) level of a difference in travel time between the two routes for a 6 AM departure? Carry out an appropriate test. Show test statistic and p-value.
 - Is there statistically significant evidence (at the 0.05) level of a difference in travel time between the two routes for a 7 AM departure? Carry out an appropriate test. Show test statistic and p-value.
 - Is there statistically significant evidence (at the 0.05) level of a difference in travel time between the two routes for an 8 AM departure? Carry out an appropriate test. Show test statistic and p-value.
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4. This data exists on the number of hurricanes that have hit Typhoon Island, a place that lies on the equator and thus has hurricanes throughout the year.

NUMBER OF HURRICANES HITTING TYPHOON ISLAND

Year	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016
Number	2	5	7	1	4	0	9	3	8	5

- A. Assume that the number of hurricanes each year is independent with a common standard deviation, but that the intensity can differ randomly from year to year. Then derive a large sample estimate for the expected number of hurricanes in 2017 and the standard deviation of this estimate.
- B. Assume that monetary damages from each hurricane is independently distributed including being independent of the number of hurricanes that occur in one year. Use the information on monetary damages from 8 randomly chosen hurricanes below along with your answer to A above to give an estimate for the expected total monetary damages from all hurricanes in 2017

MONETARY DAMAGES FROM 8 RANDOMLY CHOSEN HURRICAINES

Hurricane	1	2	3	4	5	6	7	8
Damages	5	6	4	8	7	6	5	6